

Hongdao Meng

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EDUCATION

New York University

Master of Science in Computer Science GPA: 3.8

Sep. 2024 - May 2026 (Expected)

New York, NY

Beijing University of Technology

Bachelor of Engineering in Information Security GPA: 4.0

Sep. 2020 - Jul. 2024

Beijing, CN

SKILLS

Languages: Java, Python, C/C++, Go, SQL, JavaScript, TypeScript, HTML/CSS, Shell, PHP, MATLAB, LaTeX

Frameworks: React, Angular, Vue.js, Django, Flask, Node.js, Spring Boot, PyTorch, TensorFlow, Pandas, Scikit-Learn

DataBase: MySQL, Redis, MongoDB, MilvusDB, PostgreSQL, DynamoDB, Oracle, Firebase, RocketMQ, Elasticsearch

Tools: Git, Docker, AWS, Azure, CMake, Postman, CI/CD, Jenkins, Nginx

WORK EXPERIENCE

Software Engineer Intern @ C2SMARTER Center, New York

Jan. 2025 - Present

- Developed a full-stack RAG-based chatbot system using **React**, **Redux**, and **Flask**, delivering responsive UI and seamless communication, **improving user engagement by 33.7%** and **reducing data retrieval latency by 23.6%**.
- Built a scalable back-end infrastructure with **Docker**, **MongoDB**, **MilvusDB**, and **RESTful APIs**, enabling efficient data storage and retrieval, which improved query performance by **25.6%** and reduced deployment setup time by **15.7%**.
- Deployed application on **AWS EC2** and implemented **CI/CD** pipelines with **Jenkins**, ensuring high availability and automated testing, which decreased deployment cycles by **18.7%** and supported concurrent users from **5k to 10k**.
- Implemented monitoring and observability with **Grafana** and **Prometheus**, providing real-time system insights and **reducing mean time to resolution (MTTR) by 25.3%**.
- Streamlined development and quality assurance processes with **Postman** for API testing and **GitLab** for version control, resolving 82% of integration issues pre-deployment and **improving team collaboration efficiency by 22%**.

Software Engineer Intern @ DeepFake Detection Startup, New York

Sep. 2024 - Dec. 2024

- Developed core modules of deepfake detection web platform using **React** and **TypeScript** for seamless real-time interaction, enabling **1,200+** concurrent users and **reducing client-side rendering latency by 21.3%**.
- Built real-time communication layer using **Django** and **WebSocket** for robust middleware communication, delivering real-time updates with **<180ms** latency, **improving user task efficiency by 25.6%**.
- Deployed backend services on **Kubernetes (AWS EKS)** with **ELB** load balancing and **HPA** policies.
- Optimized **PostgreSQL** query execution through composite index tuning, **reducing average response time by 18%**.

Software Engineer Intern @ QingTeng, Product R&D Department

Feb. 2024 - Aug. 2024

- Developed a real-time IoT analytics platform using **Django** and **Kafka**, processing **720k+** sensor events/day with 80ms P95 latency, enabling predictive maintenance alerts.
- Configured and deployed **Nginx** as a reverse proxy for both the analytics platform and the full-stack dashboard, optimizing request routing and enhancing system security.
- Automated **AWS** resource provisioning using **Terraform** and managed configuration with **Ansible**, cutting environment setup time from **4 hrs to 50 min** across 3 regions.
- Designed a robust **Kafka** cluster with cross-AZ replication (factor=3), ensuring **99.95% availability** and reliable data ingestion for both the real-time analytics and dashboard functionalities.

Software Engineer Intern @ Data Mining & Security Lab, Beijing

Sep. 2022 - Jul. 2024

- Developed a real-time threat detection platform using **Spring Boot** and **MyBatis**, processing 150k+ security logs/day with optimized batch SQL queries, reducing alert latency by 22% (**from 650ms to 500ms P95**).
- Designed **RabbitMQ**-based notification system with priority queues, **achieving <120ms latency for critical alerts (top 5% events) and 25% higher throughput**.
- Implemented **Redisson** distributed cache with LRU eviction policy, achieving a **26% reduction in database load** and ensuring consistency across services.
- Deployed **ELK Stack (Elasticsearch, Logstash, Kibana)** for centralized logging and real-time analytics, **enhancing debugging efficiency and cutting error resolution times by 35%**.
- Designed and optimized the database schema in **MySQL** to handle millions of transactions daily, **reducing query latency by 25.8%** and ensuring data integrity in a high-concurrency environment.